PHILIP ROBERT BAUMANN

philiprbaumann@gmail.com • (908) 477 1375 • www.philiprbaumann.com

TECHNICAL SKILLS & AWARDS

Languages: C#, Python, Java, SQL, KQL, Bicep, Ruby, Golang. **Technologies:** Azure, AKS, Docker, Networks, gRPC.
Awards: *Peer Recognition Award (2025); Engineering Excellence Award (2023), People's Choice Culture Awardee (2021).*

EXPERIENCE

MICROSOFT

Senior Software Engineer, Tech Lead, Security Copilot New York, NY
Dec 2025 – Present

- Leading a 5→8 engineer team as Architect and Tech Lead for AI Evaluation and Observability on Security Copilot; driving cross-org requirements gathering across 3-4 Microsoft Security teams and translating competing constraints into a phased execution roadmap amid 100x user growth.
- Architected and drove development of an E2E telemetry SDK and ingestion pipeline for Security Copilot, handling platform-wide customer content and model training logs at ~1.2 GB/s across 20+ regions with automated provisioning, transit guarantees, PII redaction, and compliance constraints.
- Designing a distributed AI evaluation platform for Security Copilot with strict ordering across concurrent agent flows; supports pluggable scoring including LLM-as-judge, latency, token consumption, and user-defined metrics with configurable thresholds.

Senior Software Engineer, Security Copilot New York, NY
Nov 2023 – Present

- Created an AI evaluation and reporting system that served as the quality backbone of Security Copilot. Partnered cross-org to transition and scale the architecture into Azure AI Foundry's infrastructure which now serves 70k+ customers including ~80% of Fortune 500 businesses.
- Defined AI evaluation strategy for Microsoft Security, establishing standardized quality metrics, review processes, and tooling adopted cross-org. Presented strategy and learnings to VP/CVP leadership to shape investment priorities and cement long-term evaluation roadmap.
- Launched a compliant human-in-the-loop labeling platform; onboarded 100+ partners and produced 20k+ labels in 1 year, reducing model upgrade iteration cycles from weeks to hours.
- Developed Markov Chain graph generators to optimize generative AI task selection by leveraging probabilistic skill routing to increase user engagement by ~15% (published on arXiv 2506.20815).

Software Engineer II, Security Copilot New York, NY
Apr 2023 – Nov 2023

- Built an automated quality measurement service to analyze 10k+ daily AI responses across 7 signals to improve Security Copilot; instituted weekly quality reviews adopted cross-functionally.
- Mentored an intern and led tasks for 3 junior engineers; intern converted to full-time.

Software Engineer II, PromoteIQ New York, NY
Dec 2021 – Apr 2023

- Key contributor to ultra-low-latency (<100ms) ad server generating $3M+ daily revenue.
- Reduced Azure spend by ~$60k a month via a Java utility for pre-aggregated metric generation.
- Built hourly multi-threaded file distribution system using Azure DevOps with 99.99% uptime.
- Developed high load, low latency MVP for the execution of untrusted ad bidding logic on sensitive user data to replace 3P cookies; coordinated with cross-functional and open-source communities.

IBM

Software Engineer and Development Lead, Aspera San Francisco, CA
Jul 2019 – Dec 2021

- Built 5 high-availability and Dockerized services for high-speed, large-scale data transfer.
- Scoped and developed 14+ patches and 3 official releases across 4 Ruby web applications while successfully managing IBM leadership to focus on highest priority bugs for Fortune 500 clients.
- Served as trusted representative to Fortune 250 client for monthly face-to-face discussions.
- Handpicked for IBM's Engineering Security Guild to apply industry security best practices to work.

DevOps Intern, Chief Data Office Armonk, NY
Jul 2018 – Aug 2018

- Owned construction of a secure authentication microservice for IBM's largest data lake using Java.

EDUCATION

UNIVERSITY OF NOTRE DAME, **College of Engineering, Bachelor of Science** *May 2019*

ADDITIONAL INFORMATION

Experiences: Backpacked Yosemite's Red Peak Pass; Self-taught guitarist; Built homemade solar panels; Amateur carpenter.
Interests: Growing my Digital Garden; *Dungeons & Dragons;* Strategy Games; Reading science fiction; Cooking; Running.